

# VeriPaper Test Paper: A Multi-Factor Framework

## Abstract

This paper presents a novel framework for detecting plagiarism and AI-generated content in research papers through semantic analysis and machine learning. We evaluate our approach on arXiv abstracts and benchmark datasets, achieving 85% accuracy. Our results demonstrate the effectiveness of hybrid detection methods combining embeddings with statistical features.

## 1. Introduction

Research integrity is critical to the scientific process. However, plagiarism and AI-generated content pose increasing challenges to academic publishing. In this paper we propose VeriPaper, a multi-factor assessment framework that combines semantic plagiarism detection, AI generation probability estimation, and citation validation to produce a comprehensive credibility score. The framework achieves state-of-the-art performance on multiple benchmarks.

## 2. Methods

Our approach consists of four modules. First, we extract semantic embeddings using SBERT and search a FAISS index of arXiv abstracts. Second, we measure perplexity, lexical diversity, and stopword frequency to detect AI-generated text. Third, we validate citations using the CrossRef API and check for year mismatches. Fourth, we analyze statistical anomalies including repeated decimals and unrealistic p-values. All experiments used Python 3.11 and scikit-learn 1.5.1.

## 3. Results

On a test set of 100 papers, our framework achieved 89% accuracy in identifying problematic citations, 76% precision in AI detection, and detected semantic plagiarism with 92% recall. The overall credibility score effectively distinguished authentic from suspicious papers (AUC-ROC = 0.89). The p-value for the primary comparison was  $p = 0.031$ , and the secondary outcome showed  $p < 0.001$ .

## 4. Discussion

The weighted combination of plagiarism, AI probability, and citation validity provides a transparent, explainable assessment of research integrity. While individual modules show promise, ensemble approaches improve robustness. Limitations include dependence on open corpus coverage for plagiarism search. Future work includes fine-tuning on domain-specific datasets and expanding statistical risk detection.

## 5. Conclusion

VeriPaper demonstrates that multi-factor approaches to research authenticity assessment are both feasible and effective. Our framework can support editorial decisions and help researchers improve their work. We make the code and baseline models available for community use.

## References

- [1] A. Author, B. Researcher, and C. Scientist, "Transformer architectures for document understanding," IEEE Trans. Pattern Anal., vol. 45, no. 3, pp. 112-125, 2024. doi: 10.1109/TPAMI.2024.00001.
- [2] D. Writer, "Extensive evaluation of detection metrics including precision, recall, and F1 scores," arXiv:2401.00001, 2024. doi: 10.48550/arXiv.2401.00001.
- [3] E. Analyst, "Robust performance on diverse NLP tasks and settings," in Proc. ACL, 2024. doi: 10.18653/v1/2024.acl-long.1